



Tecnológico de Monterrey

Actividad 7: NAT Individual

Alumnos:

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Fecha de entrega:

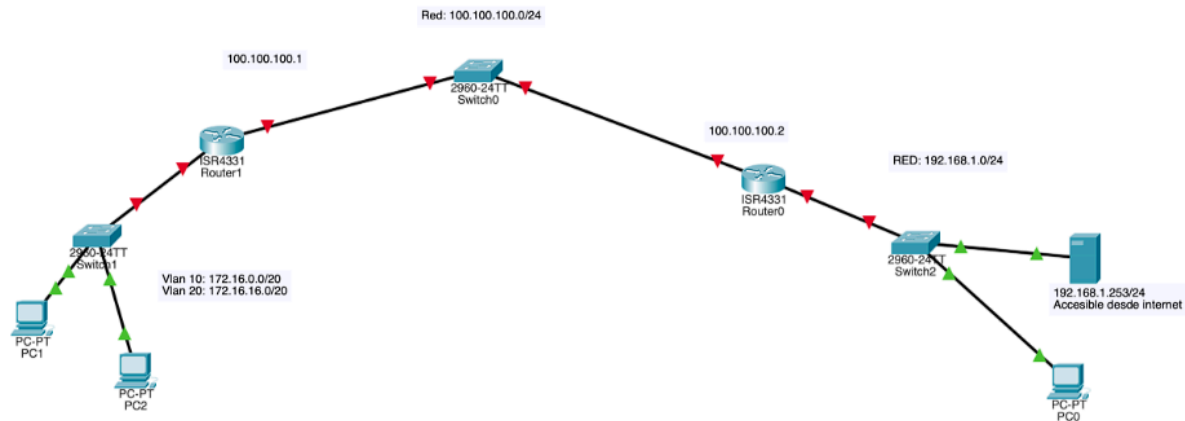
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Instrucciones:

- Realizar la siguiente topología:



- Una vez que todo esté configurado, revisa que las computadoras en la misma red se puedan hacer ping. Así mismo, las computadoras de la red del lado izquierdo deberían poder hacer ping entre redes, pero no con la red del lado izquierdo.

```
172.16.0.2
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 172.16.16.2

Pinging 172.16.16.2 with 32 bytes of data:

Reply from 172.16.16.2: bytes=32 time<1ms TTL=127
Reply from 172.16.16.2: bytes=32 time<1ms TTL=127
Reply from 172.16.16.2: bytes=32 time<1ms TTL=127
Reply from 172.16.16.2: bytes=32 time<1ms TTL=127

Ping statistics for 172.16.16.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 172.16.0.1: Destination host unreachable.
Reply from 172.16.0.1: Destination host unreachable.
Request timed out.
Reply from 172.16.0.1: Destination host unreachable.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- En el router del lado izquierdo, configura NAT para que todas las redes tengan salida a internet.

None

```
interface GigabitEthernet0/0/1.10
ip nat inside
exit
```

```
interface GigabitEthernet0/0/1.20
ip nat inside
exit
```

```
interface GigabitEthernet0/0/0
ip nat outside
exit
```

```
access-list 1 permit 172.16.0.0 0.0.31.255
ip nat inside source list 1 interface GigabitEthernet0/0/0
overload
```

- En el router del lado derecho, configura NAT para que todas las computadoras tengan acceso a internet y además, este el puerto 80 del servidor con la ip 192.168.1.253 expuesto a internet.

None

```
interface gigabitEthernet 0/0/1
ip nat inside
exit
```

```
interface gigabitEthernet 0/0/0
ip nat outside
exit
```

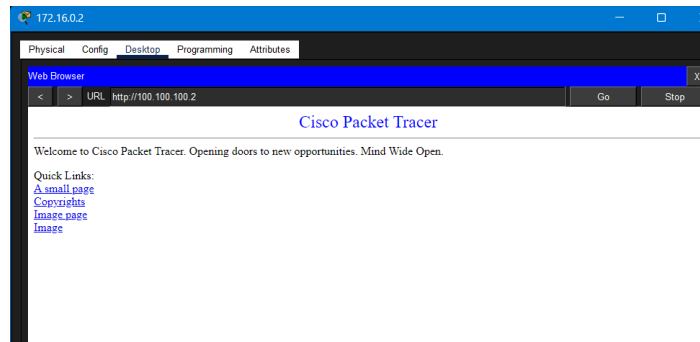
```
access-list 1 permit 192.168.1.0 0.0.0.255
```

```
ip nat inside source list 1 interface
```

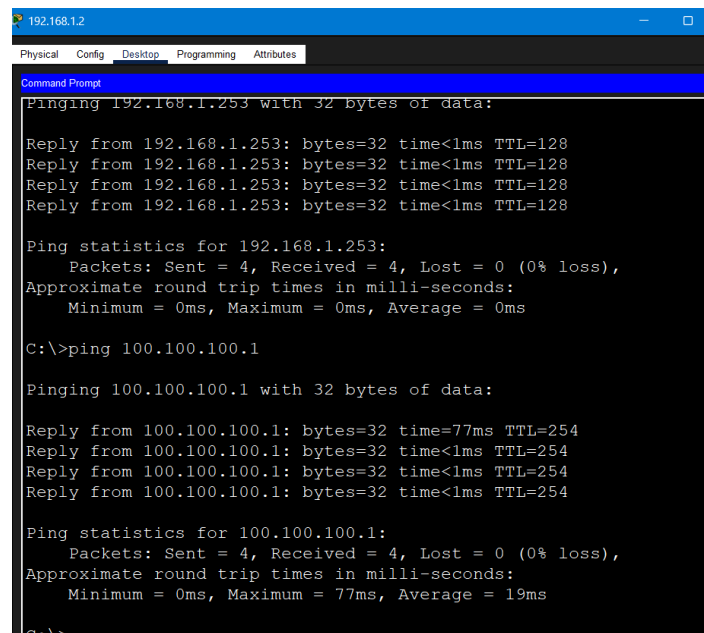
```
ip nat inside source static tcp 192.168.1.253 80
100.100.100.2 80
```

- Al final, las computadoras del lado izquierdo deben poder acceder al servidor web del lado derecho. Así mismo, la computadora del lado derecho debe poder hacer ping al router 100.100.100.1.

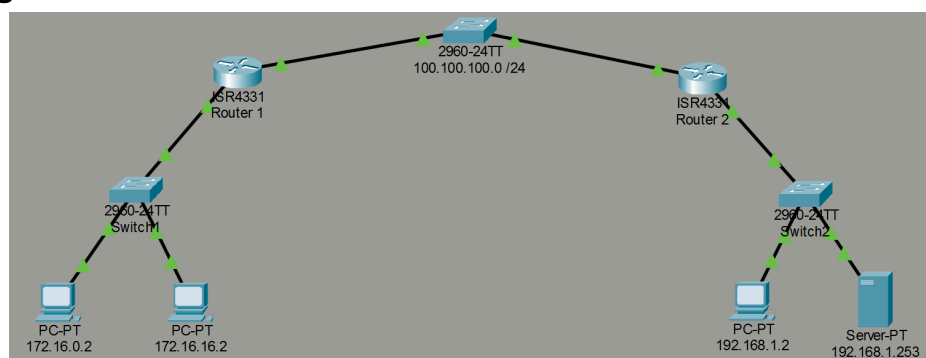
Computadora 172.16.0.2 entrando al servidor web del lado derecho:



Computadora 192.168.1.2 haciendo ping a 100.100.100.1:



Topología final:



Configuraciones finales:

Router 1:

None

Current configuration : 1029 bytes

!

version 16.6.4

no service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname Router_1

!

ip cef

no ipv6 cef

!

spanning-tree mode pvst

!

interface GigabitEthernet0/0/0

ip address 100.100.100.1 255.255.255.0

ip nat outside

duplex auto

speed auto

!

interface GigabitEthernet0/0/1.10

encapsulation dot1Q 10

ip address 172.16.0.1 255.255.240.0

ip nat inside

!

interface GigabitEthernet0/0/1.20

encapsulation dot1Q 20

ip address 172.16.16.1 255.255.240.0

ip nat inside

!

interface Vlan1

no ip address

!

```
ip nat inside source list 1 interface GigabitEthernet0/0/0
overload
ip classless
ip route 0.0.0.0 0.0.0.0 100.100.100.2
!
ip flow-export version 9
!
access-list 1 permit 172.16.0.0 0.0.31.255
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
end
```

Switch1

None

Current configuration : 1205 bytes

```
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport access vlan 10
switchport mode access
!
```

```
interface FastEthernet0/2
switchport access vlan 20
switchport mode access
!
interface GigabitEthernet0/1
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
end
```

Router 2:

```
None
Current configuration : 909 bytes
!
version 16.6.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router_2
!
ip cef
no ipv6 cef
!
spanning-tree mode pvst
!
```

```
interface GigabitEthernet0/0/0
ip address 100.100.100.2 255.255.255.0
ip nat outside
duplex auto
speed auto
!
interface GigabitEthernet0/0/1
ip address 192.168.1.1 255.255.255.0
ip nat inside
duplex auto
speed auto
!
interface Vlan1
no ip address
!
ip nat inside source list 1 interface GigabitEthernet0/0/0
overload
ip nat inside source static tcp 192.168.1.253 80
100.100.100.2 80
ip classless
ip route 0.0.0.0 0.0.0.0 100.100.100.1
!
ip flow-export version 9
!
access-list 1 permit 192.168.1.0 0.0.0.255
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
end
```